Predicting the Electric Output of Combined Cycle Power Plants Using Ensemble Machine Learning Methods

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Abstract

Predicting the electric output of combined cycle power plants is critical for optimizing energy management and maximizing the efficiency of converting fossil fuels into electricity. This study explores the application of ensemble machine learning methods to forecast net hourly electric output using ambient variables such as temperature and pressure. Using data collected from a Combined Cycle Power Plant operating at full load between 2006 and 2011, an ensemble framework combining ridge regression, random forest, support vector regression, and k-nearest-neighbors was developed and compared to a meta-learner utilizing lasso regression to predict future plant electrical output based on ambient conditions. The model achieved an R-Squared value of 95.5% and a Root Mean Squared Error (RMSE) of 3.23. These findings demonstrate the potential of ensemble learning to improve predictive accuracy in power plant operations, offering valuable insights into key factors that influence performance and help plant operators maintain optimal operating conditions.

1 Introduction

The demand for efficient and sustainable energy production has become increasingly critical in the face of growing global energy needs and environmental concerns. Combined cycle power plants (CCPPs), which utilize both gas and steam turbines to generate electricity, are a cornerstone of modern energy production due to their high efficiency and reduced greenhouse gas emissions compared to traditional fossil fuel power plants (SOURCE). Combined cycle power plants utilize excess heat from gas turbines to power steam turbines, providing higher fuel efficiency. In a gas turbine, natural gas is mixed with air and ignited, spinning a generator. A combined cycle power plant captures exhaust heat from this process through a Heat Recovery Steam Generator (HRSG), which generates steam that is then delivered to a steam turbine. This process allows combined cycle power plants to be up to 50% more efficient than a traditional power plant (https://www.gevernova.com/gas-power/resources/education/combined-cycle-power-plants). However, their performance is significantly influenced by ambient conditions such as temperature, atmospheric pressure, and humidity, making accurate prediction of power output a vital task. Effective forecasting of electric output is essential for optimizing energy management, and ensuring the economic and environmental sustainability of power plants. It is both economically and environmentally advantageous to utilize the potential energy in fossil fuels more effectively.

Traditional methods of performance analysis, often based on thermodynamic modeling, can be computationally intensive and less adaptable to varying environmental conditions and factors. Machine learning methods are capable of addressing these limitations by offering scalable, data driven solutions which can adapt to complex patterns in large or varied data sets. Power plants can analyze electrical output based on their own in-situ measurements, providing greater insights into optimal operating conditions at a specific power plant.

This study aims to address the challenge of accurately predicting the net hourly electric output of CCPPs by utilizing an ensemble machine learning approach. The objectives of this research are threefold: (1) to develop and evaluate an ensemble framework that combines multiple regression techniques for improved predictive accuracy, (2) to analyze the importance of ambient variables in influencing plant performance, and (3) to provide actionable insights that can help operators optimize plant operations under varying conditions.

By leveraging ensemble learning methods, this study seeks to advance predictive modeling in energy systems and contribute to the broader adoption of data-driven approaches in the energy sector.

2 Exploratory Data Analysis

The scope of this work includes using data collected from a CCPP operating at full load between 2006 and 2011, which provides a comprehensive dataset for model training and evaluation. Average conditions are recorded from sensors around the power plant that record ambient variables every second, and directly influence the thermodynamic efficiency of the power plant. The features consist of hourly average ambient variables:

Feature	Range
Ambient Temperature (T)	1.81 °C to 37.1125 °C
Exhaust Vacuum (V)	25.36 to 81.56 cm Hg
Ambient Pressure (AP)	992.89 to 1033.30 milibar
Relative Humidity (RH)	25.36% to 100.16%
Net Hourly Electrical Output (PE)	420.26 to 495.76 MW

The dataset was examined for missing or inconsistent values, but no significant data cleaning was required.

Here is the scatterplot matrix for all features and target variable:

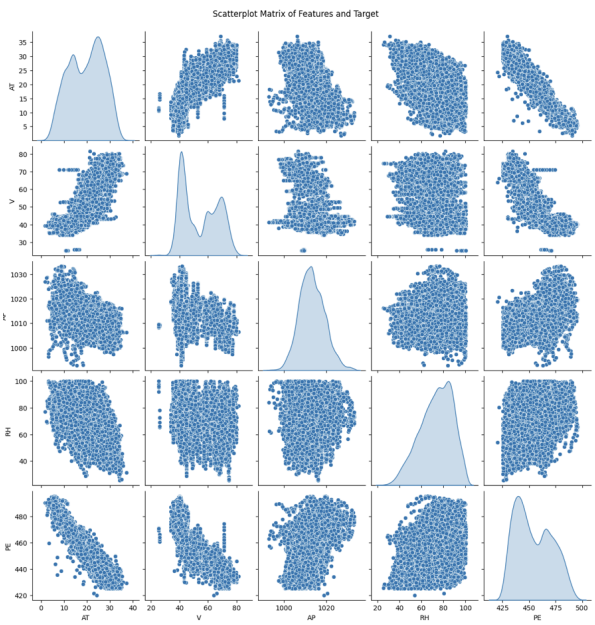


Figure 1: Scatterplot Matrix of Features.

Analysis:
The histogram suggests a **bimodal distribution** with two prominent peaks around 440 and 470. This implies that the electrical output clusters around two primary values rather than being normally distributed. This could indicate that the power plant operates in two distinct modes or under different conditions, possibly due to variations in demand, input variables (e.g., temperature, pressure), or operational settings.
Further investigation of features such as ambient temperature, pressure, or humidity is needed to understand what causes the bimodal behavior.
Similarly, variables such as **Exhaust Vacuum** and **Ambient Temperature** also exhibit bimodal distributions. Net Hourly Electrical Output and Ambient Temperature show a direct **negative linear relationship**.

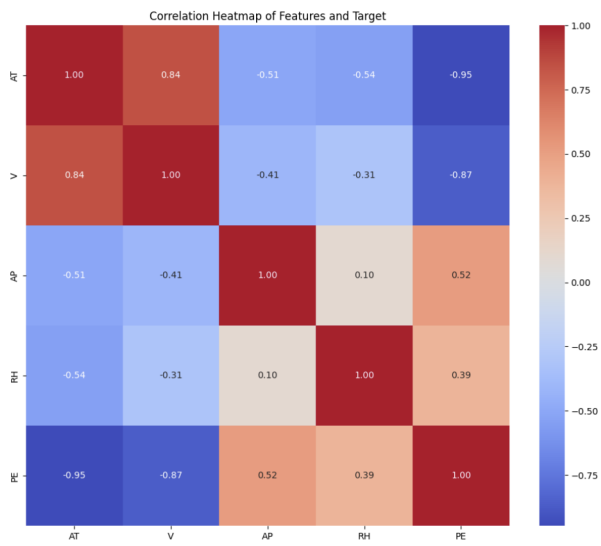


Figure 2: Correlation Heat Map.

Analysis:
The correlation heat map validates the observed relationship, showing a **-94.8% correlation** between Ambient Temperature (AT) and Net Hourly Electrical Output. This highlights that AT is inversely related to electrical output.
Additionally, the heat map suggests that **AT will likely be the most important predictor of electrical output** in future predictive models. Understanding this strong negative correlation can help refine feature selection and improve model performance.

Feature	Correlation with PE	VIF
PE	1	-
AP	0.518	66.618
RH	0.389	40.704
V	-0.869	74.969
AT	-0.948	39.157

VIF is defined as the inverse of tolerance.

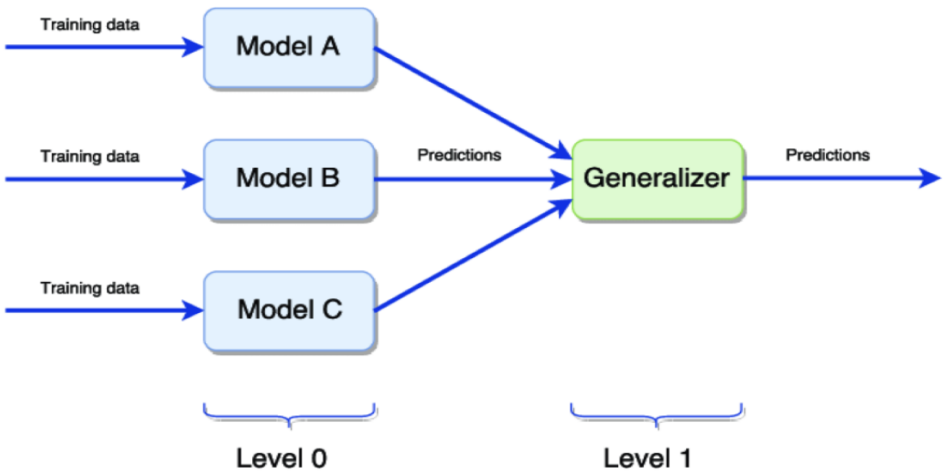
$$VIF = \frac{1}{R_i^2}$$

The Variance Inflation Factor quantifies the inflation in coefficients due to multicollinearity. $VIF = 1 \implies$ no collinearity, and $VIF > 10 \implies$ higher collinearity. All features' variance exceeds the chosen threshold of ten, indicating high correlation between features. AT was removed from the model given its high VIF and -94.8% correlation, but led to an increase in residuals due to a loss of predictive power. This is because the ambient conditions are not independent of each other, and the collinear variables collectively explain a significant portion of electrical output's variance. Removing one of the ambient conditions resulted in useful information being discarded, and less predictive capability in the model. This indicates there was unexplained trends in the data not captured by the model due to interactions between features being removed. Thus, regularization is more desirable. By regularizing, the model retains information regarding the relative influence of all features and facilitates relationships relevant under specific operating conditions.

3 Model Development

The data was split into training and testing data using an 80-20 ratio.

An ensemble learning method was utilized for this problem:



Source: https://www.analyticsvidhya.com/blog/2020/12/improve-predictive-model-score-stacking-regressor/

A stacked generalization model combines predictions from multiple base learners to improve the overall performance, and uses meta-learner to best combine the predictions of the base models. Stacking leverages the strengths of multiple models, reducing overfitting

compared to relying on a single model, making it more generalizable based on different power plants' unique data. The meta learner takes the predictions from the base model and combines these predictions optimally to minimize errors.

A preliminary model with linear regression, ridge regression, decision tree, random forest, support vector regression, k-nearest-neighbors, and a meta-learner using ridge regularization was made, but experienced overfitting due to redundancy in the selection of ensemble methods.

Table 1: Comparison of Training and Test Metrics

Metric	**Training**	**Test**
Mean Squared Error (MSE)	1.759	10.567
Root Mean Squared Error (RMSE)	1.326	3.250
R-squared (R^2)	0.993	0.963

The model significantly over performed on the training data, with a difference in R-Squared of 3% and a difference in MSE of 8.81. This was fixed by limiting the model to ridge regression, random forest regression, support vector regression, and k-nearest-neighbors. A diverse set of base learners is important, as each model can capture its own patterns in the data. Regularization within each model was added, and feature scaling for SVR and KNN, to prevent overfitting to patterns in the data.

1. Base Learners

Ridge Regression (f_1)

$$\hat{y}_1 = f_1(X) = \hat{\beta}_\lambda^{ridge} = \arg\min_\beta (y - X\beta)^T(y - X\beta) + \lambda||\beta||^2$$

The L_2 norm penalty is $||\beta||^2 = \sum_{j=1}^p \beta_j^2$. The tuning parameter $\lambda > 0$ controls the trade-off between regression fitting and coefficient shrinkage.

```
('ridge', Ridge(alpha=1.0))  # Regularized Ridge Regression
```

Listing 1: Ridge Regression Code

The regularization strength λ is set to 1.0

Random Forest (f_2)

$$\hat{y}_2 = f_2(X) = \frac{1}{T}\sum_{t=1}^{T} h_t(X),$$

where h_t are the individual decision trees in the forest and T is the total number of trees. A random forest is desirable due to its easy interpretability and automatic stepwise variable selection. However, it suffers from high variance, which is why it is balanced by other methods in the ensemble.

The random forest function in SciKitLearn does not use an impurity function to determine optimal tree splits, but uses a variance reduction technique, minimizing mean squared error, defined as

$$\Delta Var = Var_{parent} - (\frac{n_{left}}{n_{total}} + Var_{left} + \frac{n_{right}}{n_{total}} Var_{right})$$

Support Vector Regressor (SVR, f_3)

$$\hat{y}_3 = f_3(X) = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*) K(x_i, x) + b$$

where α_i, α_i^* are Lagrange multiplies, the radial basis function kernel is defined as $K(x_i, x) = exp(-\gamma ||x_i - x||^2)$. $\gamma = \frac{1}{p \cdot Var(X)}$ and controls the influence of a single training example. Additionally, the data was standardized using $x_j^{scaled} = \frac{x_j - \mu_j}{\sigma_j}$

The SVR algorithm minimizes $\frac{1}{2}||w||^2 + C \sum_{i=1}^{n} (\xi_i^+ + \xi_i^-)$ subject to the constraints $y_i - f(x_i) \leq \epsilon + \xi_i^+$ and $f(x_i) - y_i \leq \epsilon \xi_i^-$. C is the regularization parameter for this model.

k-Nearest Neighbors (KNN, f_4)

$$\hat{y}_4 = f_4(X) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(X)} y_i,$$

where $\mathcal{N}_k(X)$ is the set of indices for the k-nearest neighbors of X.

2. Meta-Learner (Lasso Regression)

The meta-learner f_m (Lasso regression) takes the predictions from the base learners as inputs. Lasso stands for Least Absolute Shrinkage and Selection Operator, which uses L_1 regularization to shrink less important coefficients to zero.

Let $X \in \mathbb{R}^{n \times 4}$ be the matrix of base learner predictions:

$$X = \begin{bmatrix} \hat{y}_1^{(1)} & \hat{y}_2^{(1)} & \hat{y}_3^{(1)} & \hat{y}_4^{(1)} \\ \hat{y}_1^{(2)} & \hat{y}_2^{(2)} & \hat{y}_3^{(2)} & \hat{y}_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ \hat{y}_1^{(n)} & \hat{y}_2^{(n)} & \hat{y}_3^{(n)} & \hat{y}_4^{(n)} \end{bmatrix},$$

where $\hat{y}_j^{(i)}$ is the prediction of the j-th base model for the i-th sample.

Lasso regression is trained on X to minimize:

$$\hat{\beta}^{lasso} = \arg\min_{\beta} (y - X\beta)^T (y - X\beta) + \lambda ||\beta||_1 \text{ where } ||\beta||_1 = \sum_{j=1}^{p} |\beta|_j|$$

This provides automatic feature selection, but in its role as the meta-learner, it selects the most important base models based on their contributions to the final prediction. Lasso assigns a weight to each base model's prediction, and shrinks it to zero if it does not contribute significantly to improving the accuracy of the final prediction, and removes it from the ensemble. This means only the base models with the most useful predictions are retained.

The final stacked model prediction $\hat{y}$ is:

$$\hat{y} = f_m(X) = X\boldsymbol{\beta}_m.$$

4 Results

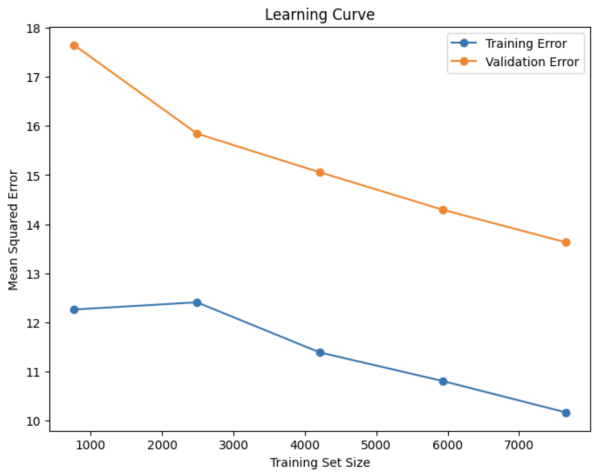


Figure 3: Validation and Training Error Over Training Set Size.

Analysis:
Training Error generally decreases as the training set size increases, which is expected because the model can better generalize with more data, and starts to stabilize with a larger training set size. Validation Error decreases as the training set size increases, indicating improved model performance on unseen data as more training data is used. However, a distinct gap between training and validation errors implies some degree of overfitting.

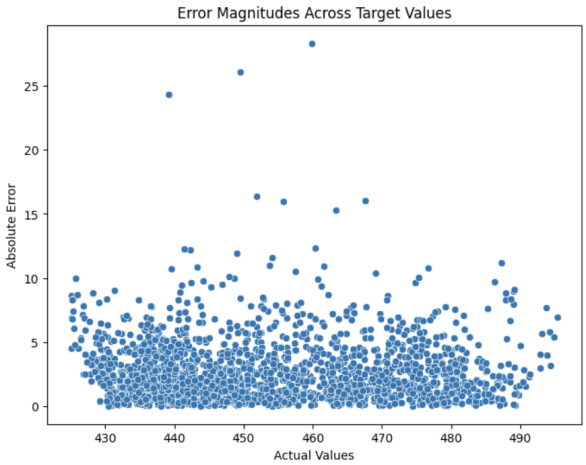
Table 2: Comparison of Training and Test Metrics

Metric	**Training**	**Test**
Mean Squared Error (MSE)	10.4691	13.0994
Root Mean Squared Error (RMSE)	3.2356	3.6193
R-squared (R^2)	0.9641	0.9548

After introducing regularization and removing redundant methods from the ensemble, the model was no longer overfitting to the training data. There was a <1% difference between the R-Squared of the training and testing data, compared to 3% before regularization. The vertical distance between training and validation error in Figure 3 is quantified in Table 2, and the difference between the metrics explains the visual gap.

GRAPH SHOWING WHICH MODELS IN THE ENSEMBLE ARE MOST/LEAST IMPORTANT:

This plot effectively visualizes the relationship between the actual target values (on the x-axis) and the corresponding absolute errors (on the y-axis). It can be a useful diagnostic plot for assessing model performance. Here are some observations and interpretations: Spread of Errors: The majority of the absolute errors appear to cluster at lower values, indicating that the model performs reasonably well for most predictions. There are some larger errors (above 20) that could indicate potential outliers, poorly predicted data points, or regions where the model struggles. Error Patterns Across Target Values: If the errors show any systematic pattern (e.g., increasing or decreasing with the actual values), it might indicate bias or heteroscedasticity (non-constant variance in errors). In this plot, the errors appear relatively evenly distributed across the range of target values, suggesting that the model does not exhibit clear bias with respect to the magnitude of the target. Model Diagnostics: A random, patternless distribution of errors (as seen here) is generally desirable, as it indicates the model is performing consistently across the target values. The presence of a few large errors might suggest areas for improvement, such as further feature engineering, adding more training data, or trying a different model. This plot is a valid and insightful way to diagnose model performance and detect any systematic issues in predictions.



Analysis:

Figure 4: Correlation Heat Map.

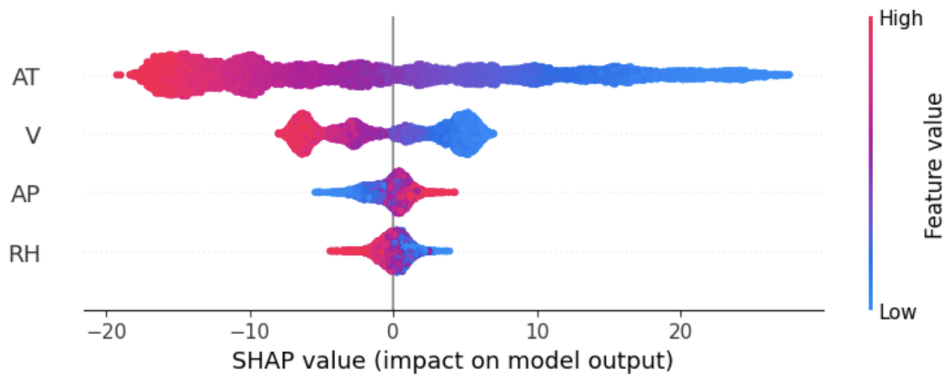
The presence of a few large errors might suggest areas for improvement, such as further feature engineering, adding more training data, or trying a different model. This plot is a valid and insightful way to diagnose model performance and detect any systematic issues in predictions.

Yes, this graph suggests a few areas you might want to explore further to ensure your model's performance is robust and optimized: 1. Investigate Large Errors The points with absolute errors greater than 20 could represent: Outliers in the data (e.g., extreme target values or noisy inputs). Model weaknesses in capturing certain patterns in the data. What to do: Examine the data points corresponding to large errors. Check if they have unusual features or if they represent edge cases. Consider removing outliers (if they are not representative of your target distribution) or applying transformations to reduce their influence. If those errors correspond to specific target ranges or feature subsets, refine your feature engineering or try specialized modeling techniques.

This is likely due to overfitting given the high collinearity and VIF scores. Therefore, cross validation is useful.

5 Discussion

Interpret the results and discuss their significance. Compare them with findings from other studies if applicable. Mention any limitations of your study.



Explain the goal of using SHAP—whether it's to enhance model transparency, validate the feature set, or identify key drivers of power output. Highlight how this helps in operational decisions, like focusing on improving efficiency under specific temperature or vacuum conditions. Briefly mention that while SHAP is computationally expensive, its interpretability justifies the effort for models that need transparency or validation. (Took 3 hours to run) SHAP values represent the impact of each feature on the model's output (Power Output in this case). A positive SHAP value indicates that the feature is pushing the prediction higher. A negative SHAP value indicates that the feature is pushing the prediction lower. The further a point is from 0, the more influence that feature has on the prediction.

Each row represents a different feature in the model. The features are ranked by importance from top to bottom: AT (Ambient Temperature) is the most influential feature. RH (Relative Humidity) is the least influential among these four.

The color gradient (red to blue) represents the feature values: Red dots indicate high feature values. Blue dots indicate low feature values.

AT (Ambient Temperature): High AT values (red) lead to negative SHAP values, reducing power output. Low AT values (blue) have a positive impact, increasing power output. V (Exhaust Vacuum): Higher V values (red) contribute positively to power output, while lower values (blue) tend to reduce it. AP (Ambient Pressure): The effect of AP is more centered around zero, indicating a weaker or more balanced impact overall, but extreme values show slight variance in SHAP values. RH (Relative Humidity): Lower RH values (blue) tend to have slightly positive impacts on predictions, while higher values (red) have small negative impacts, though the overall influence is relatively minor compared to AT or V.

AT and V are the dominant predictors, with wider SHAP value distributions and clearer color separations. RH and AP have less impact on the model's output, indicated by smaller SHAP value ranges.

Your model's performance is heavily influenced by Ambient Temperature and Exhaust Vacuum, suggesting that controlling or monitoring these variables could significantly affect power output. However, Ambient Pressure and Relative Humidity have weaker influences, potentially making them less critical for optimization or monitoring.

The plot provides a clear, visual ranking of feature importance, showing that Ambient Temperature (AT) and Exhaust Vacuum (V) have the most significant impact on power output. This insight helps stakeholders understand which variables are driving the model's predictions.

SHAP values reveal nonlinear relationships between features and the target variable. For instance, the inverse relationship between AT and power output is visible, which might not be obvious from simple coefficients or summary statistics.

Including SHAP visualizations demonstrates that your model is interpretable, an essential aspect for decision-makerswho need to trust model predictions in high-stakes environments like power plant management.

The plot can justify why certain features should be prioritized or discarded in future models. For example, Relative Humidity (RH) and Ambient Pressure (AP) show relatively low impact, which could lead to a discussion on their necessity

6 Conclusion

Summarize the main findings of your report and suggest future directions for research or practical applications.

demonstrate an understanding of the scope of your model - what is it useful for, and what are its limitations

References

List all the references cited in your report in a consistent format, such as APA or IEEE. Use the `bibliography` environment if you have a `.bib` file.

Example:

- Author1, A., & Author2, B. (Year). Title of the paper. *Journal Name*, **Volume**(Issue), pages.
- Author3, C. (Year). *Title of the Book*. Publisher.